

Introduction to Artificial Intelligence

Week 11

Agenda

- Direct v. Indirect representation
- Epistatic gene
- Partially-mapped crossover
- Examples

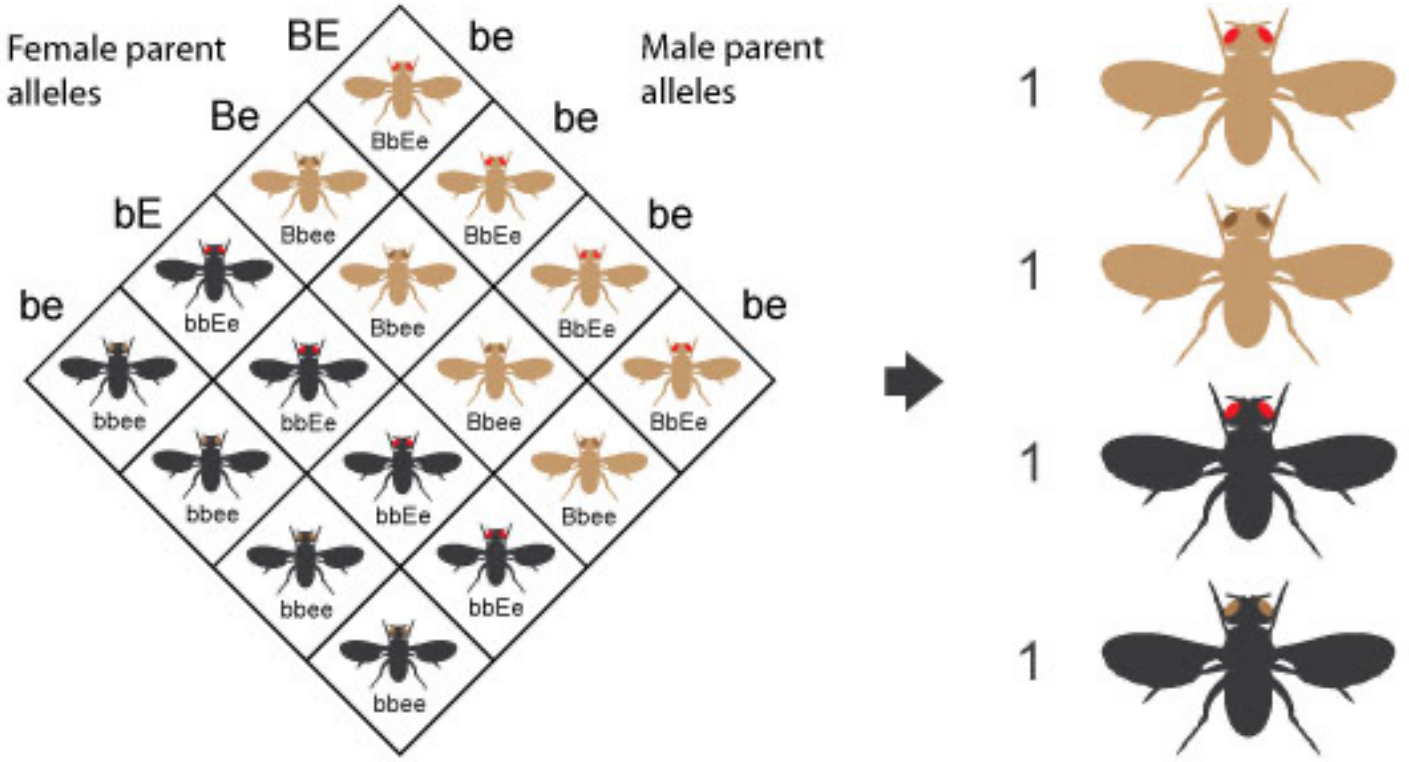
Evolutionary Algorithms

Part II

Direct v. Indirect Representations

B = Brown Body
b = Black Body

E = Red Eyes
e = Brown Eyes



- Genetics – Genotype v. Phenotype
- We may have a translation step between our internal representation in the system v. the traits perceived by someone
- The genotype of an organism is its complete set of genetic material
- Genotype contributes to phenotype, the observable traits and characteristics in an individual or organism

Direct Representation in Problems

- Direct Representation
  - When the genotype and phenotype are the same
- Examples
  - Settings on a machine
  - Creation of a FSM or other controller
- Majority of optimizations have a direct search for the outcome

Indirect Representation in Problems

- Indirect Representation
  - When the evolved values allow for the construction of a solution to the problem
- Examples
  - Evolving a program for the solution
- This is an interesting emerging method for looking at evolution of structures which leads to solutions or for the entire system

Dungeon Levels

- Genotype – a tree of tuples
- Phenotype – a board with placement of rooms
- Translation step is required

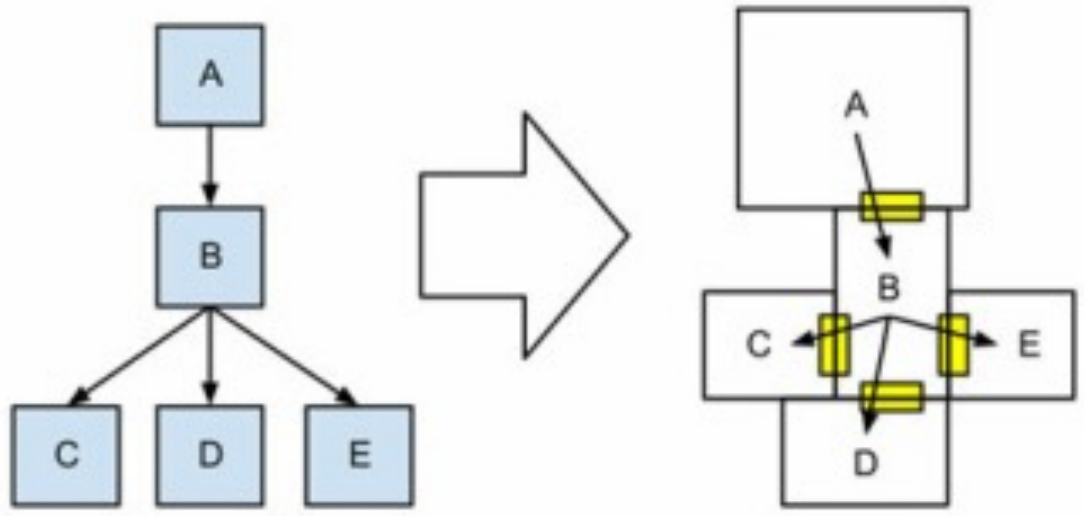


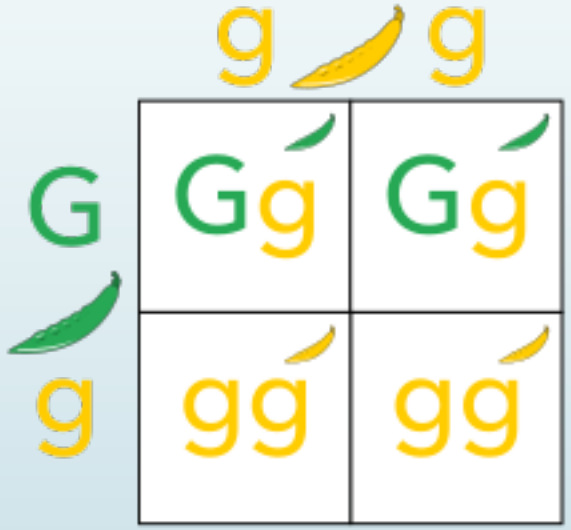
Figure 1: Example translation from tree structure into map.

Mendelian Genetics



- Discovered by **Gregor Johann Mendel** in 1865, but not popularized until Thomas Hunt Morgan in 1915
- Ratios of Pea Pods always in divisions of 1/4s
- Dominant and recessive traits
- Punnett Squares:

	R	R
r	Rr	Rr
r	Rr	Rr



What doesn't kill you makes for an interesting feature

- Sickle-cell diseases lead to a resistance to Malaria (Nature vol. 515 no. 7526)
- Серповидноклеточная анемия приводит к устойчивости к малярии

EPIDEMIOLOGY

A moving target

Migration has led to an increase in the occurrence of sickle-cell disease in countries with previously low incidence of the disorder.

BY SIMON PLEASANTS

Sickle-cell disease is a heritable blood disorder that can have devastating effects. It is the result of a minor genetic change — a single error in a single gene — yet it is one of the most common genetic causes of sickness and death[1]. It is also on the rise. According to a study published last year, approximately 305,800 children were born with sickle-cell disease in 2010, about two thirds of them in Africa[2]. By 2050, the study authors predict, those numbers will increase by more than 25%, to more than 400,000 babies born annually with the disorder.

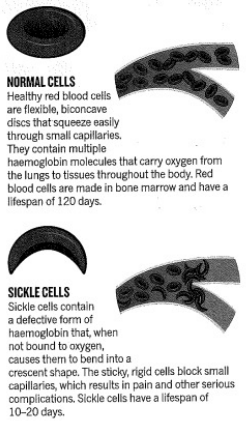
The disease-causing mutation occurs in a gene responsible for the production of haemoglobin, the protein in red blood cells that carries oxygen to tissues throughout the body. Individuals who inherit only one mutant gene typically experience no ill effects, but those who have two faulty copies develop the full-blown disease. The mutation itself is tiny: it replaces one amino acid (called glutamic acid) with another (valine) at one location in the haemoglobin protein. This small change has a major consequence — it turns the flexible, disc-shaped red blood cells rigid, giving them a distorted, sickle-like appearance.

Sickled cells are not only less pliable than healthy red blood cells, they're also stickier. This causes them to adhere to each other and plug up narrow blood vessels, reducing blood flow and preventing adequate oxygen from being delivered to the tissues that need it. This in turn leads to many complications, some of which are life threatening. People with sickle-cell disease commonly experience episodes of acute pain, known as crises, that can last from a few hours to a few days. Other complications include anaemia, leg ulcers, jaundice, kidney damage, high blood pressure, gallstones, increased susceptibility to infections and stroke. These problems arise from a variety of complex mechanisms, such as altered pain receptors (see page S8) and the decreased lifespan of red blood cells[1].

In high-income countries, improved management of the disease and medication, including strong painkillers, has meant that patients with sickle-cell disease can expect a better quality of life. In low-income nations, however, those with sickle cell are not as fortunate: in Africa, at least 50% of children born with the condition die before their fifth birthday[3].

Sickle-cell disease is most prevalent in areas with hot climates — Africa, the Indian subcontinent and the Caribbean are all affected. There is evidence to suggest that the geographic concentration of people with sickle-cell disease arose because people with sickle-cell trait — those who carry the sickle-cell gene but are generally healthy — have enhanced resistance to malaria[4], resulting in preferential selection of the gene.

The malaria-causing parasite typically invades red blood cells and multiplies inside them. But although sickle-cell carriers can be infected, the presence of sickle haemoglobin confers resistance to malaria. There are a number of different theories, one of which is that subtle biological changes in the blood — possibly higher carbon monoxide levels[5] or the interaction between oxygen and sickle haemoglobin[6]— makes people with the sickle-cell trait less of a hospitable host to the parasite and usually prevents the disease from taking hold.



NORMAL CELLS
Healthy red blood cells are flexible, biconcave discs that squeeze easily through small capillaries. They contain multiple haemoglobin molecules that carry oxygen from the lungs to tissues throughout the body. Red blood cells are made in bone marrow and have a lifespan of 120 days.

SICKLE CELLS
Sickle cells contain a defective form of haemoglobin that, when not bound to oxygen, causes them to bend into a crescent shape. The sticky, rigid cells block small capillaries, which results in pain and other serious complications. Sickle cells have a lifespan of 10–20 days.

About 70% of global cases of sickle-cell disease are in Africa. The World Health Organization estimates that in some African countries, including the Democratic Republic of Congo, Ghana and Nigeria, prevalence of sickle-cell trait is between 10% and 30%, resulting in an estimated disease prevalence of at least 2% in most of Africa — a crippling health burden in a region that is ill-equipped to bear it, and one that lacks the financial and health-care resources present in wealthy countries.

ON THE MOVE

Increased population movement associated with globalization is bringing sickle cell to regions that previously had negligibly low incidences of the disease. Migration of people with the sickle-cell trait or sickle-cell disease caused the faulty gene to flow from high-frequency sickle-cell gene areas, such as Africa and India, to western Europe, the eastern coast of South America, and North America[7]. These migrations have increased disease prevalence in areas not previously associated with the disorder. The Sickle Cell Disease Association of America estimates that there are about 100,000 people living with the disorder in the country. The US Department of Health and Human Service's National Heart, Lung and Blood Institute suggests that sickle-cell disease affects about 1 in 500 African American babies, a number about ten times greater the general population.

"In Africa, at least 50% of children born with sickle-cell disease die before their fifth birthday."

Ironically, improved medical treatment leading to longer life expectancy is contributing to the increased prevalence of sickle-cell disease. Penicillin, taken daily, helps to prevent and treat infection in young children, folic acid stimulates the production of red blood cells, the drug hydroxycarbamide (known as hydroxyurea in the United States) reduces the number and severity of painful crises, and blood transfusions help to temper acute cases of the disease. With so many interventions, more people will survive to pass on sickle-cell genes to their children. ■

Simon Pleasants *is a copy editor at* Nature.

1. Makani, J. *et al. Sci. World J.* http://dx.doi.org/10.1155/2013/193252 (2013).
2. Piel, F. B. *et al. PLoS Med.* http://dx.doi.org/10.1371/journal.pmed.1001484 (2013).
3. Grosse, S. D. *et al. Am. J. Prev. Med.* **41,** S398–S405 (2011).
4. Allison, A. C. *Br. Med. J.* **1,** 290–294 (1954).
5. Ferreira, A. *et al. Cell* **145,** 398–409 (2011).
6. Cyrklaff, M. *et al. Science* **334,** 1283–1286 (2011).
7. Piel, F. B. *et al. Lancet* **381,** 142–151 (2013).

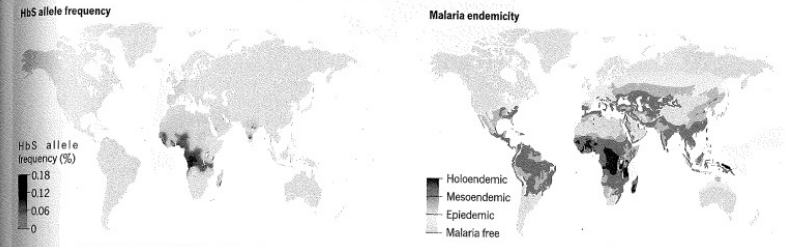
CARRIED IN THE BLOOD

Sickle-cell disease is often fatal in low-income nations. But researchers are beginning to understand more about the disease and how to treat it. By **Simon Pleasants**.

FATED LINK

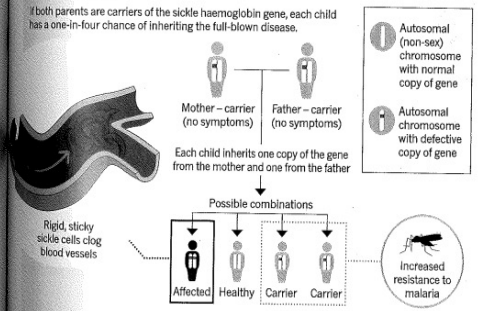
It is no coincidence that sickle-cell disease and malaria share similar global distributions. The gene responsible for sickle haemoglobin confers increased resistance to the malaria parasite when inherited from only one parent. Such a trait allowed carriers of the gene (HbS distribution, on the map below in red)[1] to survive and reproduce in areas where malaria is present (green)[2].

Although malaria is indigenous in many areas, the sickle haemoglobin gene has a more narrow distribution than malaria because the mutation arose in Africa. Limited migration from Africa to certain regions (for example, China) has restricted its spread.



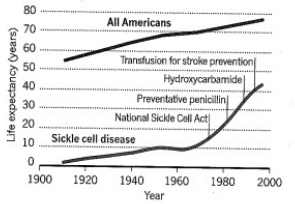
GENETIC ROULETTE

If both parents are carriers of the sickle haemoglobin gene, each child has a one-in-four chance of inheriting the full-blown disease.



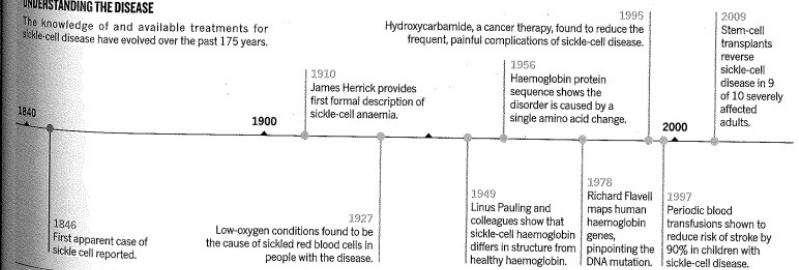
HEALTH IMPROVEMENT

As a result of more effective treatments, the life expectancy of people with sickle-cell disease in the United States has increased dramatically. Most developing nations, however, have yet to see such a change[3].



UNDERSTANDING THE DISEASE

The knowledge of and available treatments for sickle-cell disease have evolved over the past 175 years.



References: 1. Piel, F. B. *et al. Lancet* **381,** 142–151 (2013); 2. Piel, F. B. *et al. Nature Comms.* 1, 104 (2010); 3. US National Institutes of Health: National Heart, Lung and Blood Institute, available at: go.nature.com/4cuwgj.

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EA Representation

- A data structure and its defined variation operators
- Choice of representation defines the search space
  - Defines the possible solutions
    - Data structure choice
  - Defines the connection
    - Variation Operation
- How we represent a problem is a key to a good EA
- Good representations allow smaller search spaces and faster searches through the space

Self-avoiding Walks

- Take an *NxN* grid, a Self-avoiding walk of size *N-1* is a walk with *NxN* steps visiting every square once

v	>	V	>	V	X
V	^	V	^	V	^
V	^	V	^	V	^
V	^	V	^	V	^
V	^	V	^	V	^
>	^	>	^	>	^

How Can We Represent an Agent's Movement?

- Directionally – we could put each of the movement directions into a string
  - *UP, DOWN, LEFT, RIGHT* or *U, D, L, R* respectively
- The previous example would be:
  - *Down Down Down Down Down Right Up Up Up Up Up Right Down Down Down Down Down Right Up Up Up Up Up Right Down Down Down Down Down Right Up Up Up Up Up*
  - OR
  - *Down - X5, Right, Up - X5, Right, Down - X5, Right, Up - X5, Right, Down - X5, Right, Up - X5*
- The size of the search space is 4 directions raised to *N-1* steps

Can We Do Better?

D	R	D	R	D	**X**
D	U	D	U	D	U
D	U	D	U	D	U
D	U	D	U	D	U
D	U	D	U	D	U
R	U	R	U	R	U

Agent with Direction Saved

- Lets have an agent which saves its current direction it faces in
- Now movements are:
  1. *Forward (F)*
  2. *Turn Left + Forward (L)*
  3. *Turn Right + Forward (R)*
- The solution string from before is:
  - *F F F F F R R F F F F R R F F F F R R F F F F R R F F F F R R F F F F*
- Only 3 symbols raised to the *N-1*
- Large reduction in the search space
  - 3^5 = 243
  - 4^5 = 1024

What Do We Lose?

F	R	R	R	R	**X**
F	F	F	F	F	F
F	F	F	F	F	F
F	F	F	F	F	F
F	F	F	F	F	F
R	R	R	R	R	R

Search Space has been Reduced

- All legal walks still retained in the Saved Direction
- Walks where we backtrack are removed – remember this is not a movement we would be interested in
- The only walks we have reduced from the solution space where those who would never provide a correct solution – reduction of the space without negative consequence

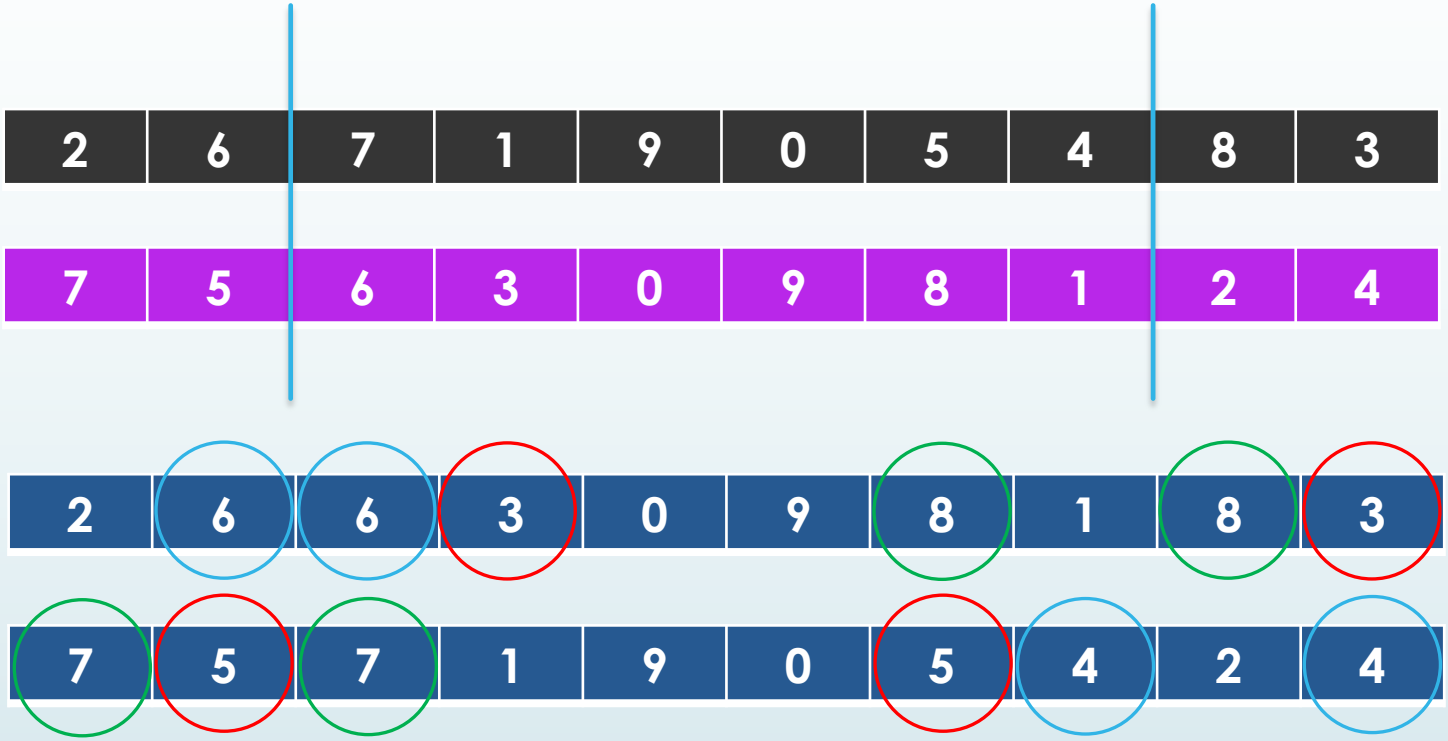
Epistasis

- **Epistatic gene** in genetics: a gene that determines whether or not a trait will be expressed
- **Epistatic gene** in EAs: the actions made in earlier parts of the chromosome having larger impacts on the fitness value of the chromosome
- **Epistatic gene** in Self-avoiding Walk: actions made early in the string will have large impacts on the fitness
- For example, changing the first move to a *R* will run the walk off the stage, giving it 0 fitness rather than 35, whereas changing the last symbol to a *R* will only reduce the fitness by 1. Assume that each correct movement gives 1 points
- *F F F F F R R F F F F R R F F F F R R F F F F R R F F F F R R F F F F F* -> *f(x) = 35*
- ***(F->R)*** *F F F F R R F F F F R R F F F F R R F F F F R R F F F F R R F F F F F* -> *f(x) = 0*
- *F F F F F R R F F F F R R F F F F R R F F F F R R F F F F R R F F F F* ***(F->R)*** -> *f(x) = 34*

TSP-Ordered Gene

- GA's representation is the tour of cities
- Natural idea of ordering in gene

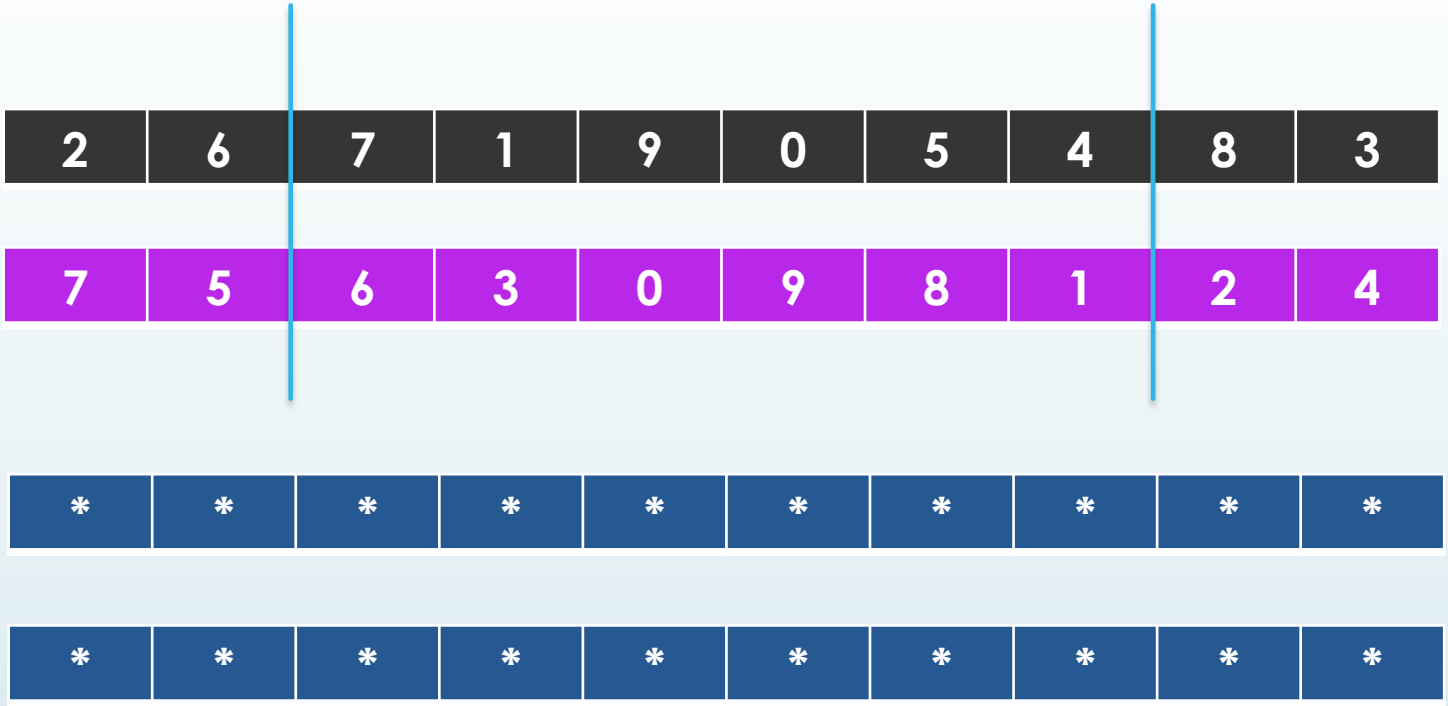
Two Point Crossover



Two parent strings

Two children strings

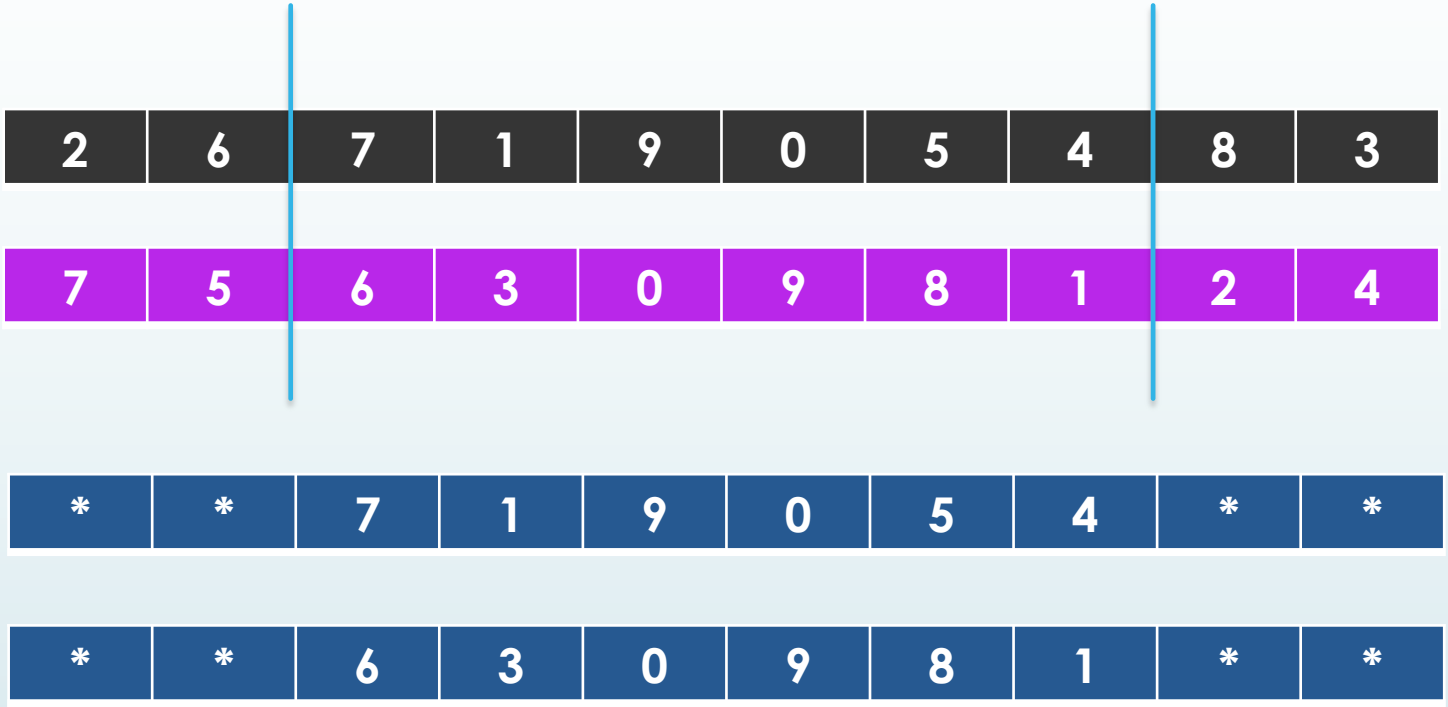
Partially-Mapped Crossover



Two parent strings

Two children strings

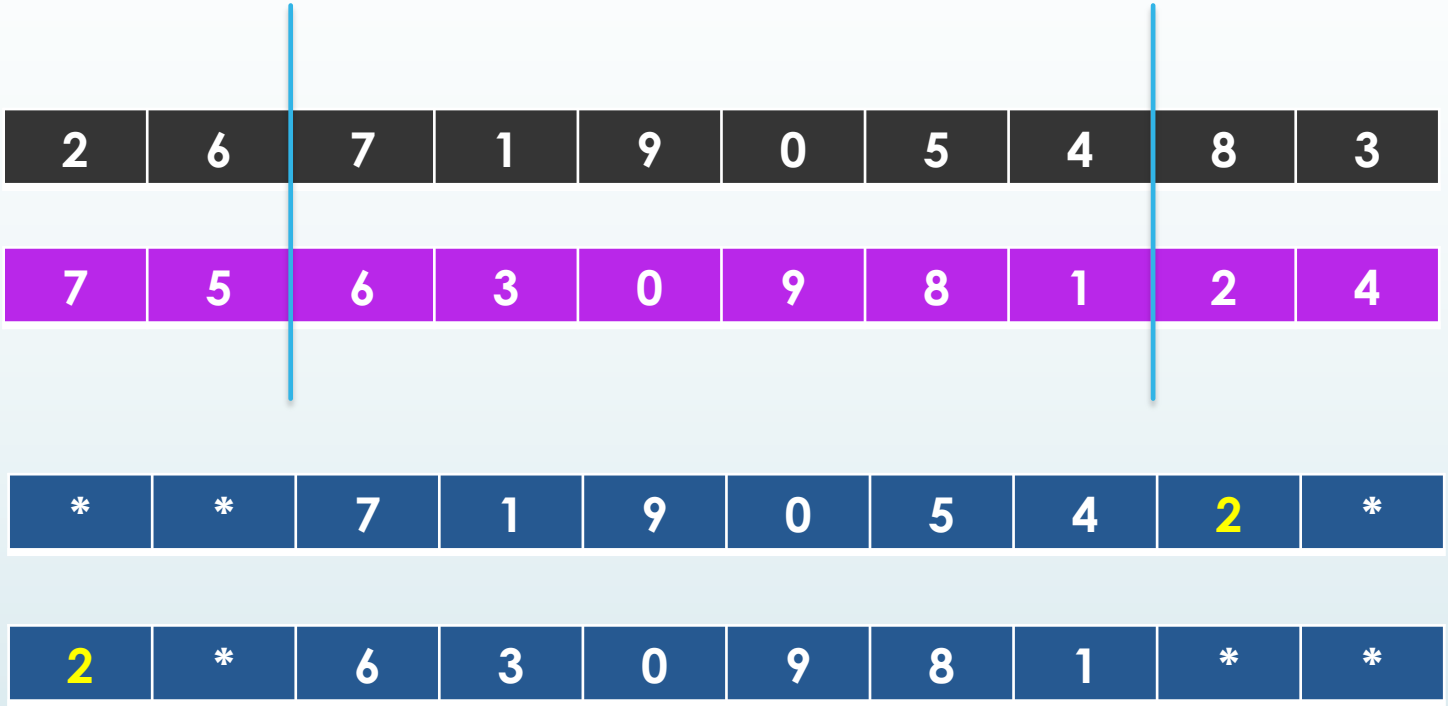
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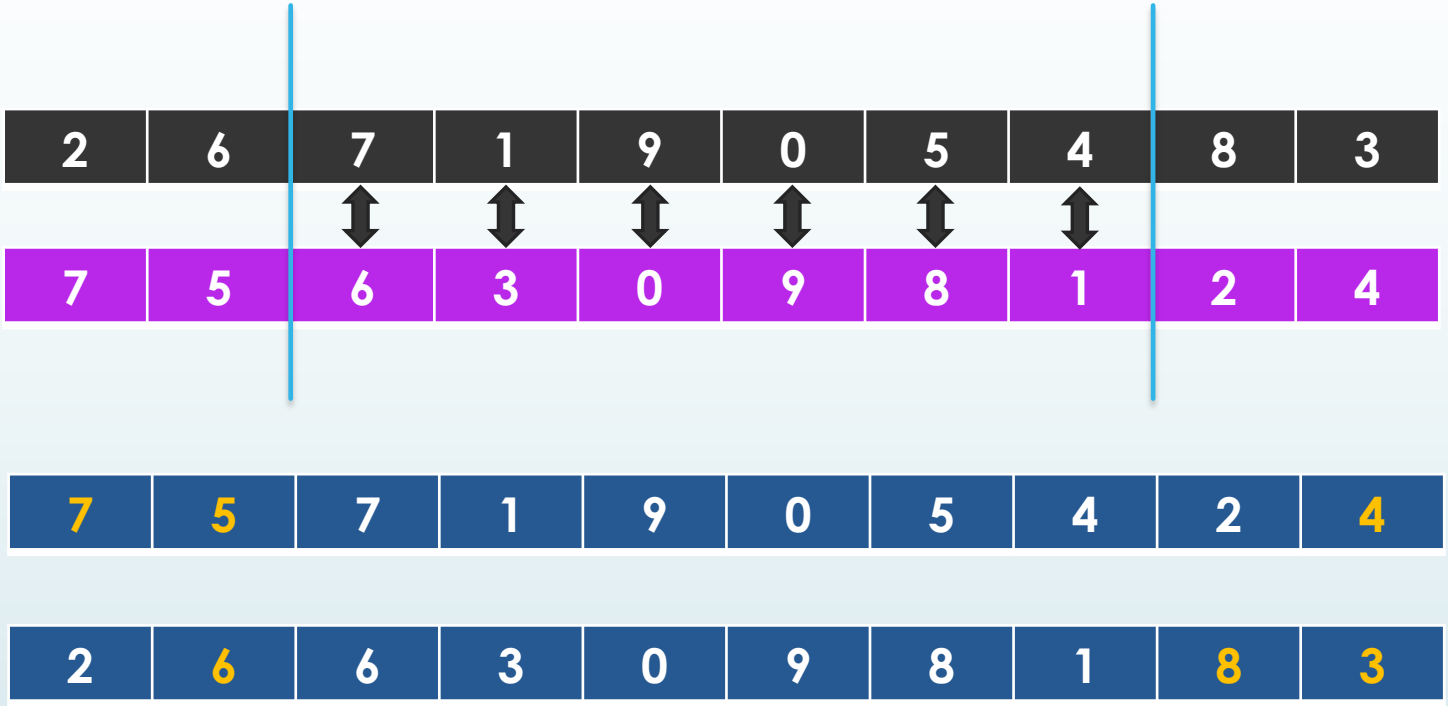
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Two parent strings

Two children strings

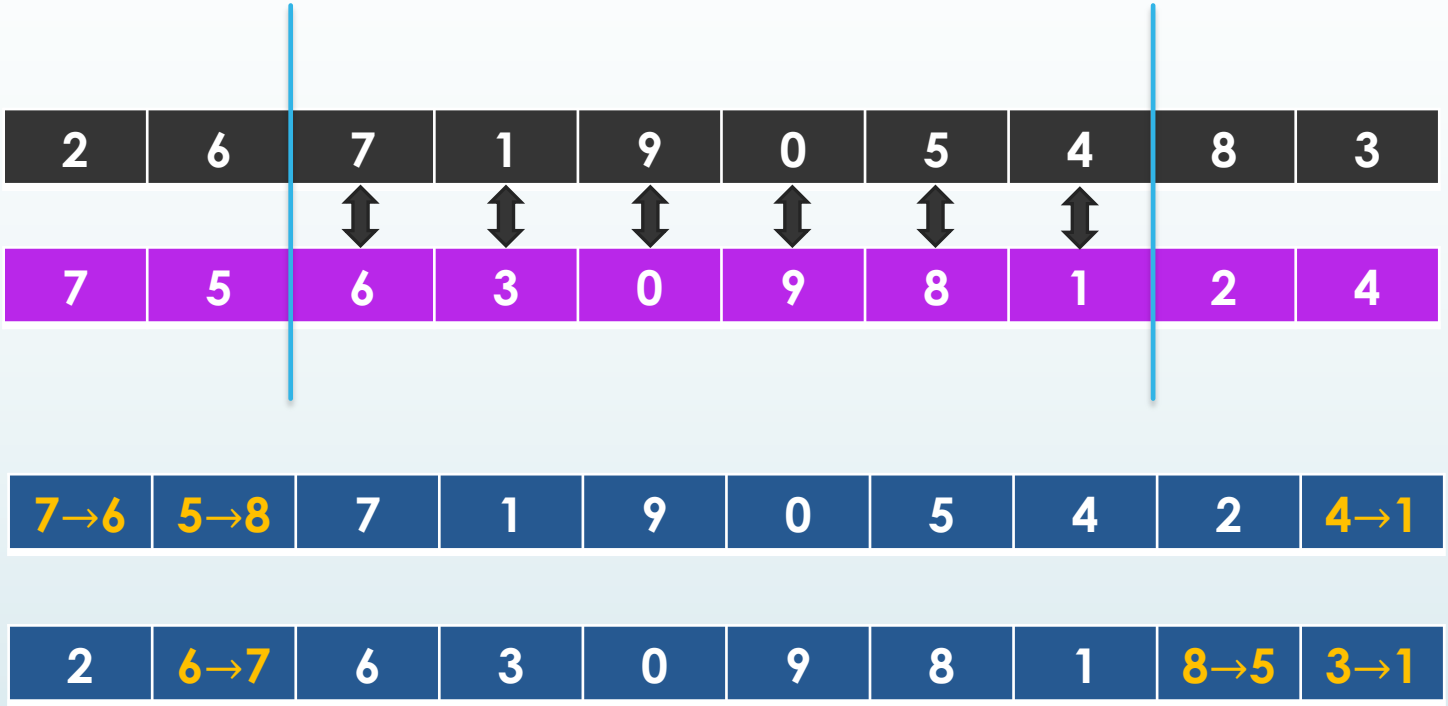
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Two parent strings

Two children strings

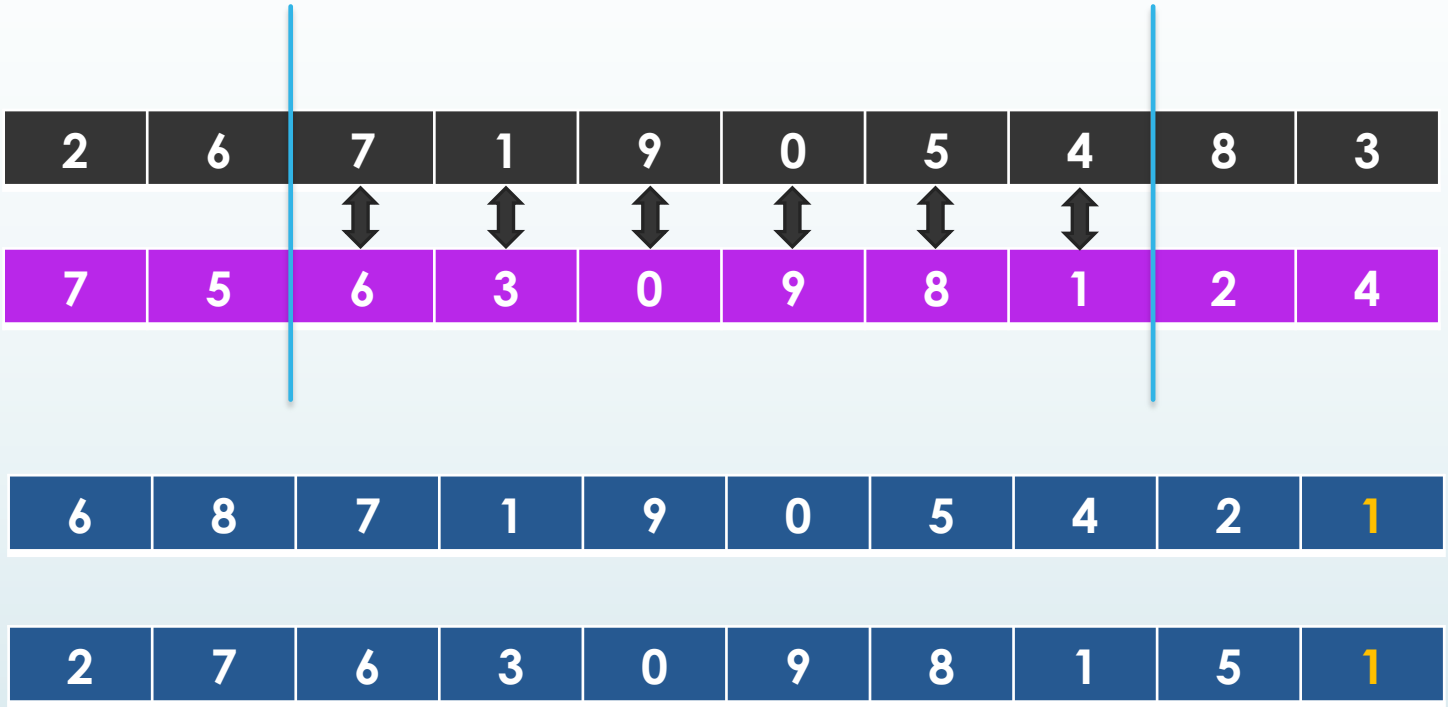
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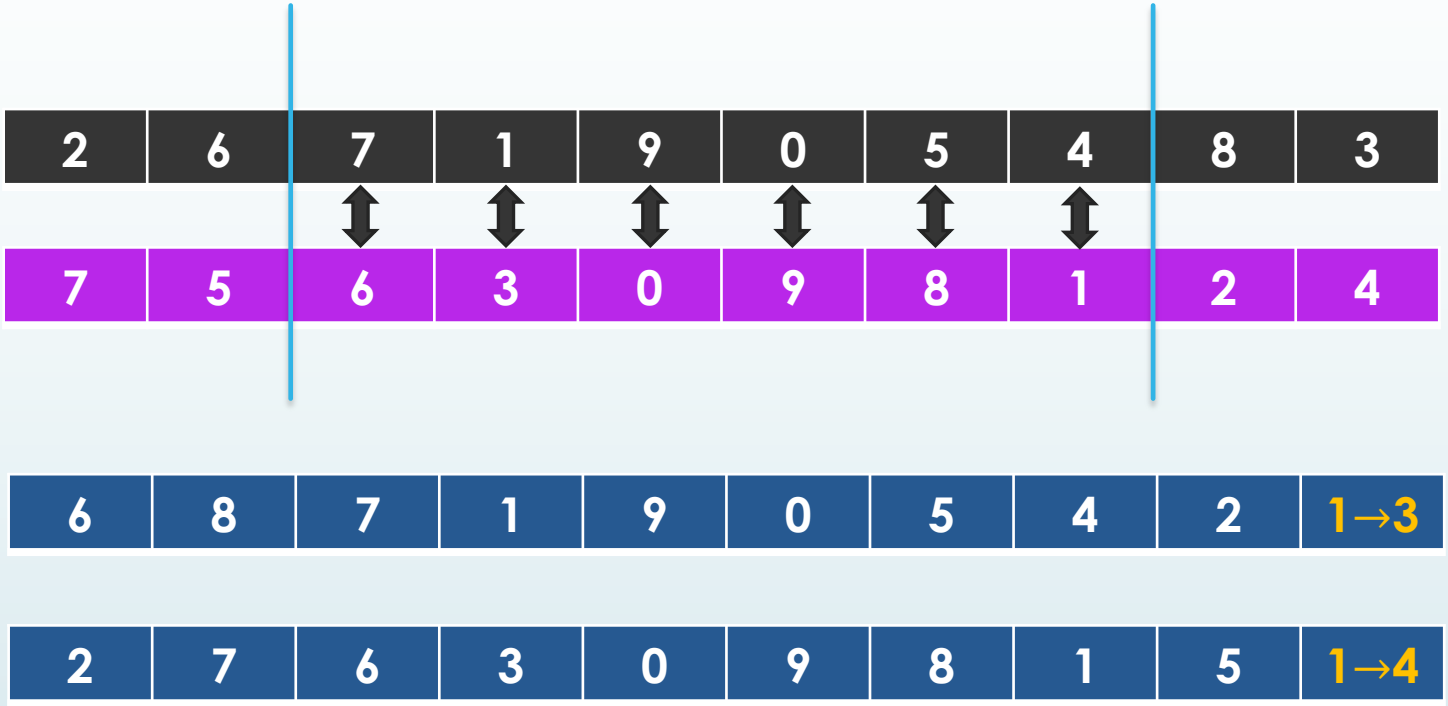
Partially-Mapped Crossover



Two parent strings

Two children strings

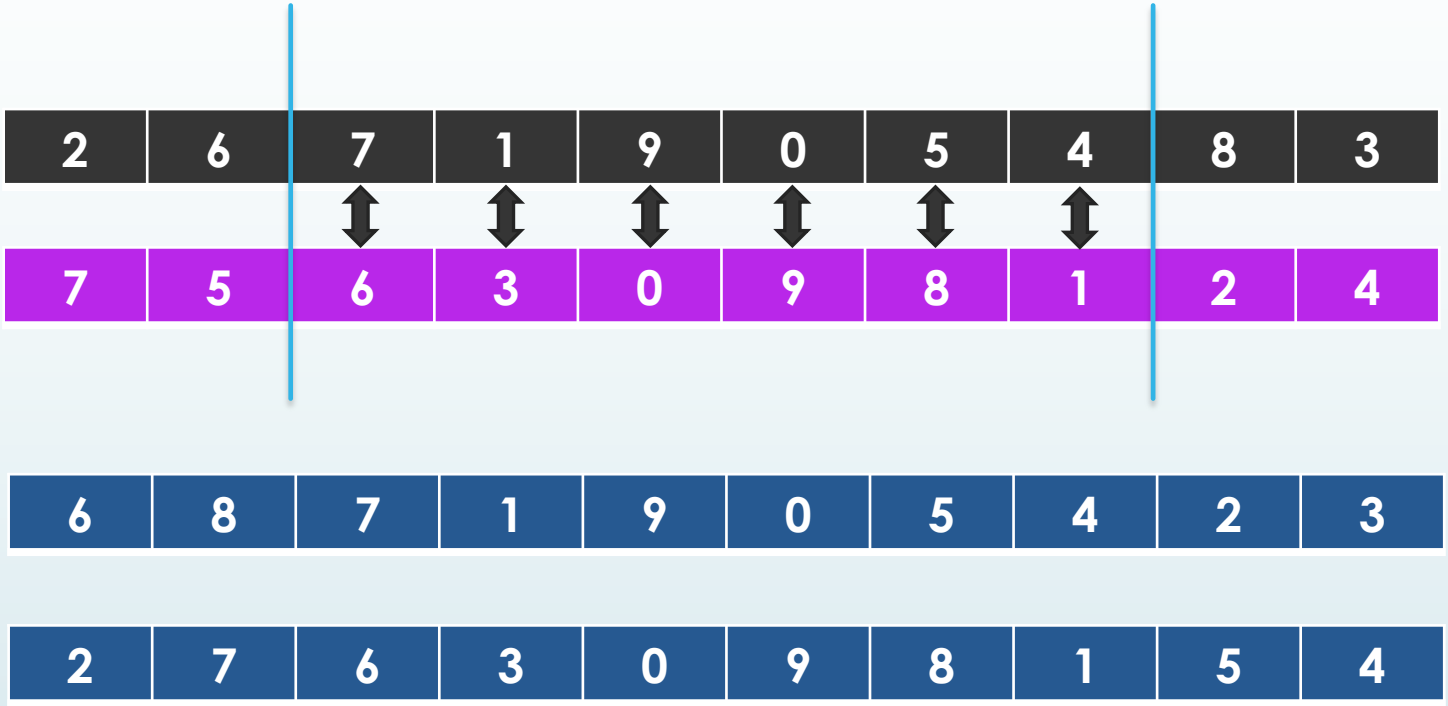
Partially-Mapped Crossover



Two parent strings

Two children strings

Partially-Mapped Crossover



Two parent strings

Two children strings

Indirect Real-Numbered Representation

- Move from direct to indirect representation
- Real-valued string
- Crossover as per one and two point
- Mutation is now as per ES – add a random number from a distribution
- Mapping form
- Example of increasing numbers

0	1	2	3	4	5	6	7	8	9
34.3	53.1	22.1	65.5	34.5	12.4	0.7	66.3	96.2	0.12

- 9, 6, 5, 2, 0, 4, 1, 3, 7, 8.

Indirect Real-Numbered Representation

- Direct to Indirect method
- No longer need to worry about ordering
  - Ordering is now taken care of in the list
  - Sorting algorithms are fast
- Evolutionary Changes are less likely to numbers which are closer together
  - Evolution also has a direct control on the heritability

Selection of Representation

- Has a large impact on the outcome of the evolution
- Defines how your space will look – the search space in terms of the connectivity on the space
- Genetic operations are defined by how they act upon the representation of the problem

Iterated Prisoner's Dilemma (IPD)

- Two player game
- Story of two convicts who are placed in interrogation and played against each other
- Exploitable player allows a score >3 against it

		Player 1: C	Player 1: D
Player 2	C	(3,3)	(5,0)
	D	(0,5)	(1,1)

Iterated Prisoner's Dilemma

- Game playing agents
- Representations
  - Function Stack Machines – FST
  - Cellular Encoded FSM – CAT
  - Finite State Machines – AUT
  - Boolean Formulas – TRE
  - Markov Chains – MKV
  - Look-up Tables – LKT
  - IF-ACTION-SKIP Lists - ISC
  - Boolean Formula with Delays - DEL

Co-evolution

- The fitness of the population is based not upon the target, but the other members within the population
- Used to create game player agents
- For IPD
  - Play each of the other players in the population once
  - Score is what you make off of all the other members in the population

Cooperative Ability

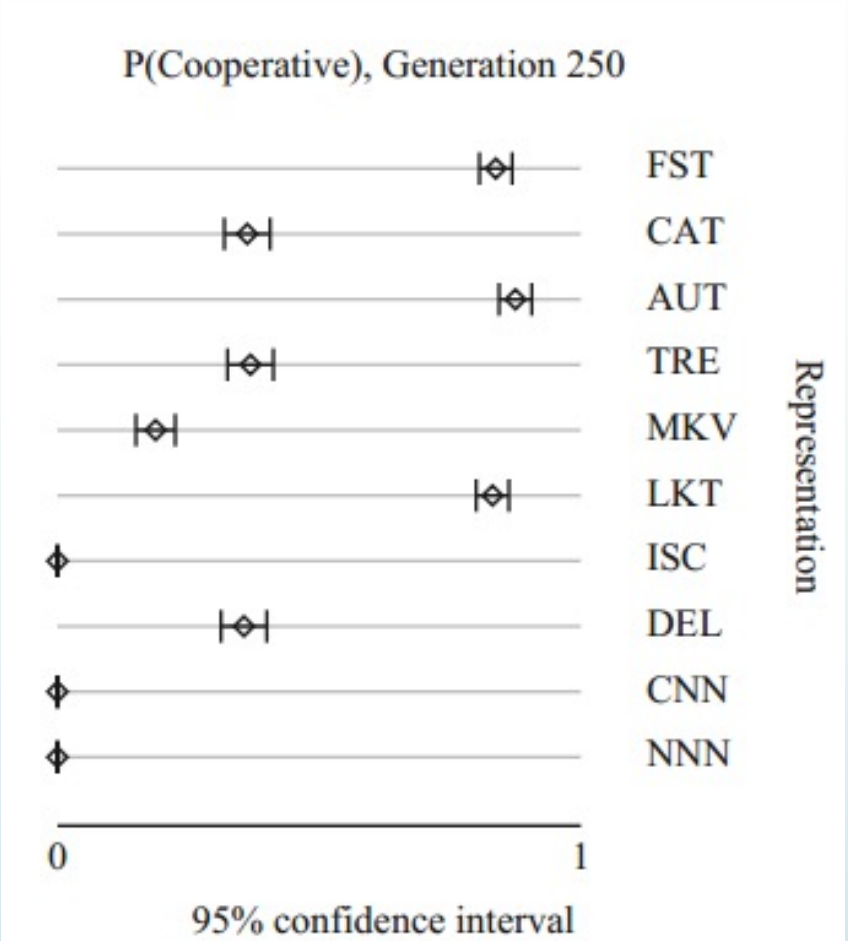


Fig. 8. Probabilities of essentially cooperative behavior in generation 250 for each of the 10 representations tested.

Play in the Population v. Random Player

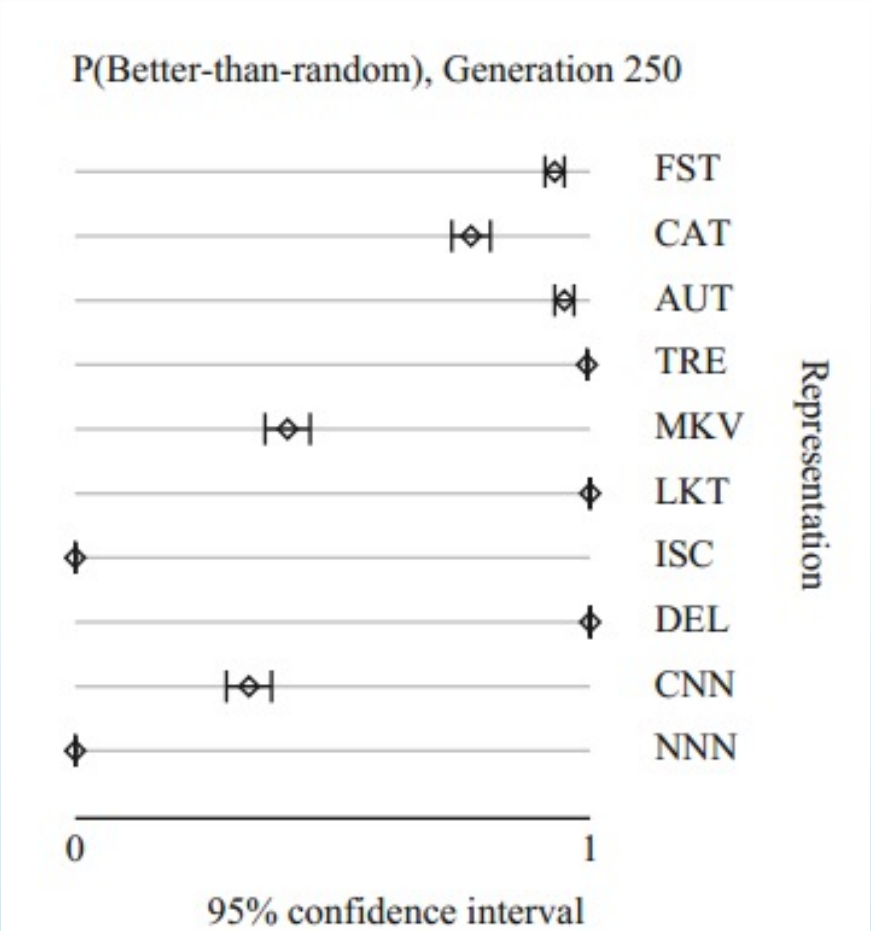


Fig. 9. Probabilities of better-than-random behavior in generation 250 for each of the 10 representations tested.

Why?

1. Initial Population

   Random initialization of all of these techniques has some differences in the ability to create an early cooperator

2. Variation Operators with Different Probabilities of Creating "Stupid" Offspring

   May allow for offspring which are easy to exploit

3. Fidelity of Reproduction

   Differences in the ability to represent ones parents. Some reproduction rules lead to drastically different children than parents

4. Granularity of Reproduction

   Cooperation must be valued to be sustained, if newer models are allowed in which can be exploited it allows for exploiters to overrun the population

String Players

- String of actions
- Actions made in a step-wise order or are representations of the player
- Have been used for a number of simulation ideas:
  - Sunburn – Space Ships
  - VIPs – Virtual Politician Simulator
  - Lessons to a young man courting – Dating Simulator
  - Dawn of the Dead - Government Responses to Zombies
- Co-evolution and Evolution to a target
  - Sunburn – co-evolved
  - VIPs – co-evolved
  - Lessons – Based on a model of the potential date
  - Dawn of the Dead – based on an epidemic model

Sunburn

- Space Warfare Simulation
- Student Project by John Walker
- Ships are represented by a string which is an indirect method of building a starships
- Ships compete in a gladiatorial tournament selection method
- Ships have a distance measure from each other – 1 dimension

Sunburn Ship

- Length N string of characters
- (S)hields – Take a hit of damage before dropping
- Weapons
  - (G)uns – Short Range Physical Projectiles which drop in effectiveness as distance increases. Slow moving and able to dodge
  - (L)asers – Can be equally effective at any range
  - (M)issiles – Guided missiles which take some time in order to ensure a lock. They have a better long range effectiveness as they track their targets
- (D)rives – allow a ship to move one space
- String kept in order – Shields are moved to front of string and tripled. Each system is worth one point of damage
- Ships have a preferred distance

Weapon Effectiveness

- A weapon hitting does one system worth of damage
- A damaged system cannot be used
- Weapons are probabilistic in their effect
- Hit based upon the weapon effectiveness curve – a function based on the current range
  - PG(r) = 0.6 - (r - 1)/38
  - PL(r) = 0.35
  - PM(r) = 0.1 + (r - 1)/38

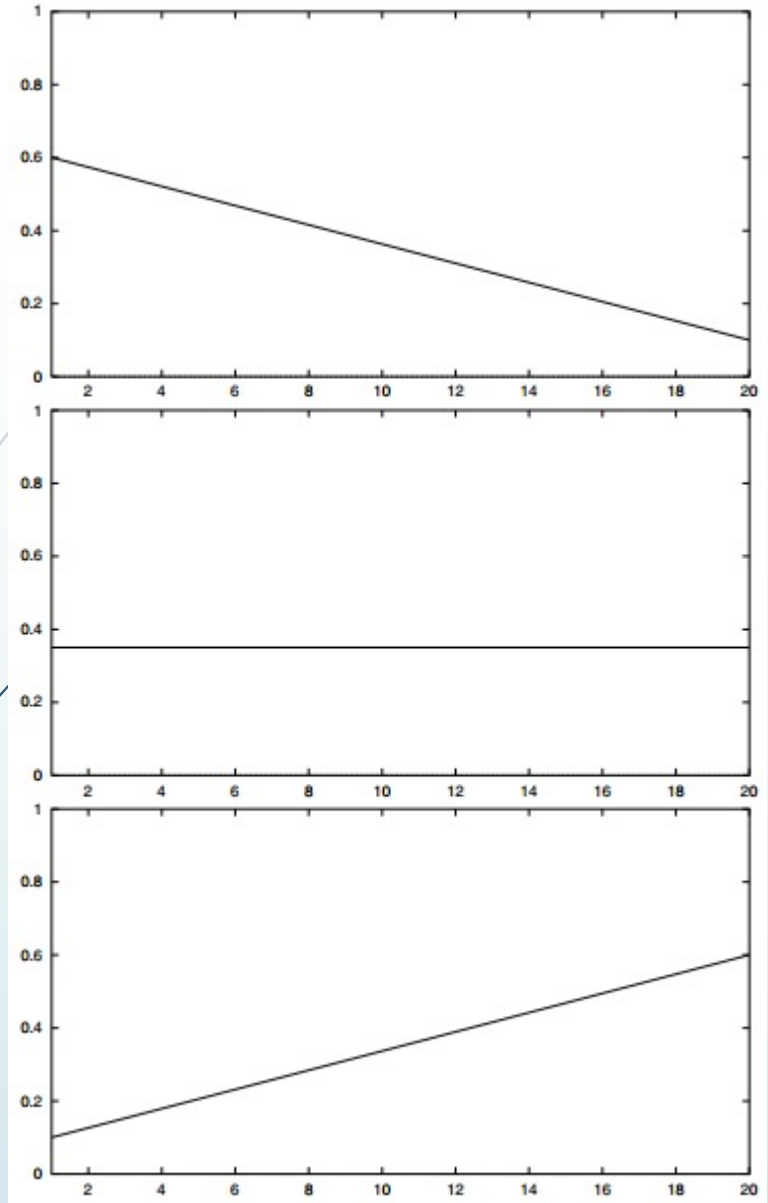


Fig. 4.3. From the top, gun, laser, and missile weapons' effectiveness curves.







Example Ship Translation

- SGLMSDSMGSLD,16
- Systems with shields in the chromosome
- Translation moves these to the front and triples them
- 4 shield nodes buys you 12 shields
- SSSS SSSS SSSS GLMDMGLD,16

Ship Controllers

- Ships activate their drives in unison one at a time with the goal of moving to their key range
- Ships fire weapons one at a time in unison, based-on their order in the string

  LLGML

  FIRE! - LASER, LASER, GUN, MISSILE, LASER

Gladiatorial Selection



Gladiatorial Fights

- Steady State Population

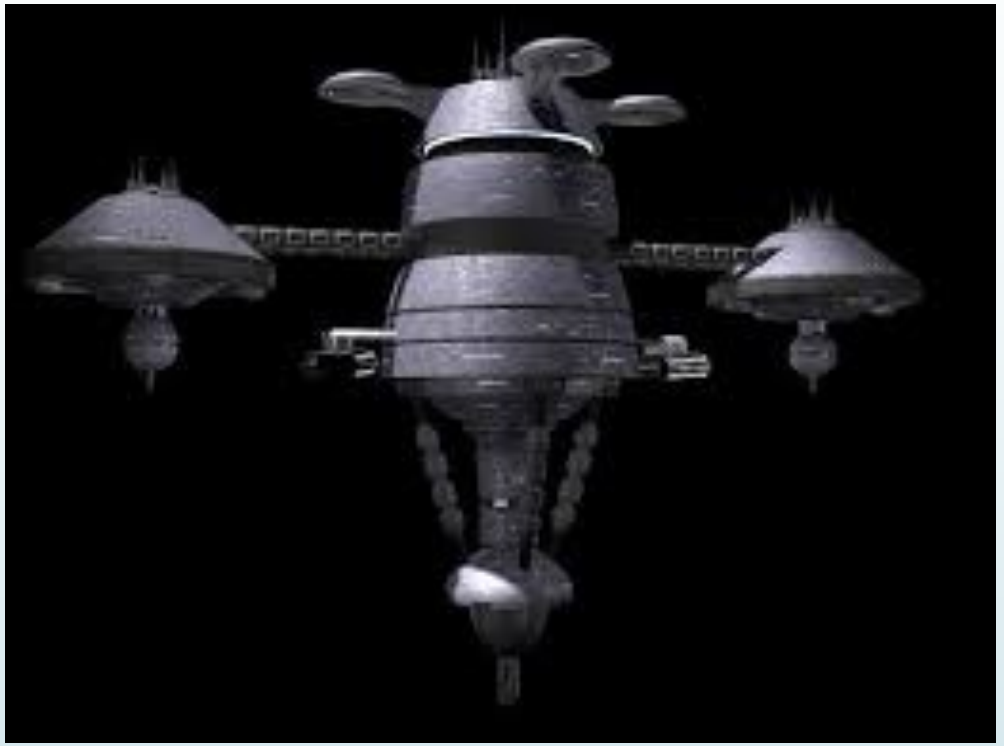
1. Pull out two ships from the population
2. Simulate their battle
3. If there is a winner – keep for breeding
4. Else return to step 1
5. Pull out two ships from the population
6. Simulate their battle
7. If there is a winner – keep for breeding
8. Else return to step 5
9. Breeding Step

Win Conditions

- A ship without any points remaining is destroyed
- 100 Turns without damage - Draw
- Are there any others?

Starbases

- Strip down to one drive in the final position
- Load up on Missiles and Shields



Gunrunners

- Lots of drives
- MORE GUN





Representational Issues

- Position within the chromosome has a large effect on the range and the damage
- Systems ‘go down’ in order
  - Expendable systems at front
  - Important systems near end
  - Win condition is a disabling of drives – ensure at least one drive is at the end of the string
- Epistatic – with importance on the end of the string

Sunburn

- What crossovers would make sense?
- What crossovers would disrupt structure?
- What about the preferred distance?

Summary

- Direct v. Indirect representation
- Epistatic gene
- Partially-mapped crossover
- Examples

Next week: ANN